Prediction of organ specific protein-protein interaction network of *Arabidopsis halleri* and identification of key genes and pathways

BBCC2021 – online edition
http://www.bbcc-meetings.it

Sayyeda Hira Hassan[a], Gabriella Sferra[a*], Melissa Simiele[a], Gabriella S. Scippa[a], Domenico Morabito[b], and Dalila Trupiano[a*]

[a]Department of Biosciences and Territory, University of Molise, 86090 Pesche, Italy

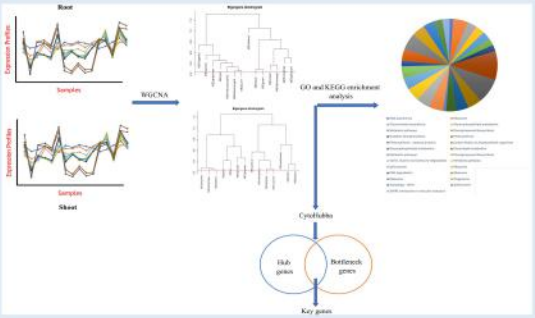
[b]Laboratoire de Biologie des Ligneux et des Grandes Cultures (LBLGC-EA1207), Université d'Orléans, 45067 Orléans CEDEX 2, France

*Corresponding authors: gabriella.sferra@unimol.it; dalila.trupiano@unimol.it

INTRODUCTION

Arabidopsis halleri is a hyperaccumulating pseudo-metallophyte and an emerging model to explore metal tolerance and hyperaccumulation (Honjo and Kudoh 2019). Protein-protein interactions (PPIs) regulate several biological functions in response to multiple signals. Thus, there is a need to identify PPIs in order to have an idea of how proteins interact in a coordinated manner to accomplish organ specific cellular activities or give response under certain circumstances, including metal stress (Senkler et al. 2017). Several experimental methods are able to identify PPIs but are costly and labor intensive (Ding and Kihara, 2019). However, analysis of high throughput data by computational methods has the potential to overcome these shortcomings and, specifically, RNA-seq is one of the main data sources used to construct PPI network (Karimizadeh et al. 2019). In this regard, the weighted gene co-expression network analysis (WGCNA; Langfelder & Horvath, 2008) has been identified as an effective biological approach helpful in identifying co-expressed modules with highly correlated genes. By considering all these aspects, the present study was aimed to predict the PPI network for *A. halleri* from RNA-seq data (Corso et al. 2021) of root and shoot by applying WGCNA and to perform a network-based identification of genes relevant to the biology of the organism by applying Cytohubba (Chin et al., 2014).

METHODOLOGY



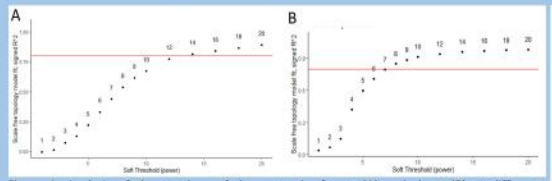


Figure 1: Analysis of the topology of the network of root (A) and shoot (B) at different thresholding powers represented by numbers in the plot.

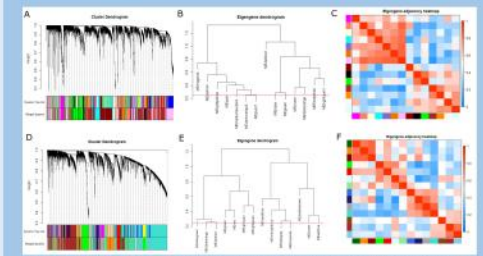


Figure 2: Weighted gene correlation network analysis in root (A-C) and shoot (D-F). Colored bars are the original and merged modules.

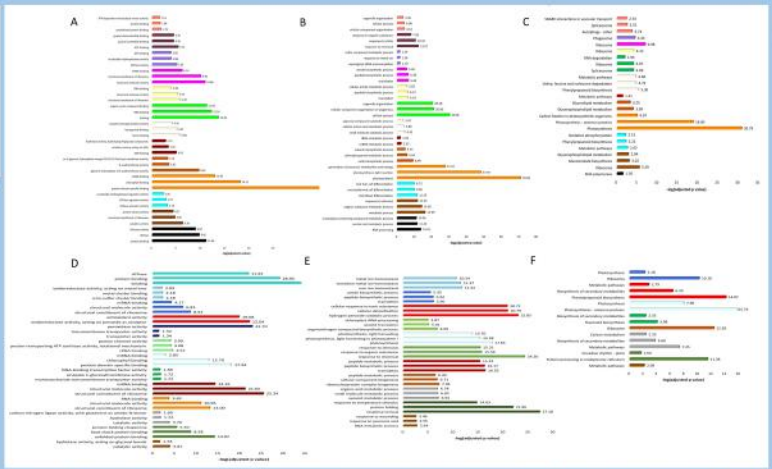


Figure 3: Bars are the three most representative molecular functions, biological processes and biological pathways enriched in each module of *A. halleri* root (A-C) and shoot (D-F). Bar colors respect module colors.

RESULTS AND DISCUSSION

We categorized 19,653 genes of root and 18,081 genes of shoot into 14 modules (groups of highly connected genes) by using soft threshold power of $\beta = 14$ for root, and $\beta = 8$ in case of shoot based on scale-free topology with $R2 = 0.97$ (Figures 1 and 2). Top 3 molecular functions, biological processes and KEGG pathways significantly enriched ($P < 0.05$) and identified in each module of root and shoot are displayed in Figure 3. The overlap among biological pathways identified in root and shoot modules are shown in figure 4. The five pathways (22.7%) common to root and shoot include photosynthesis, ribosome, metabolic pathways, phenylpropanoid biosynthesis, photosynthesis and antenna proteins. Some pathways were only identified as enriched in root including glucosinolate biosynthesis, autophagy and SNARE interactions in vesicular transport, while circadian rhythm, for example, was unique to shoot. We also identified hub and bottleneck genes of each module. Furthermore, floralwhite module showed to be enriched of key genes most of which are annotated as hypothetical/predicted proteins into public databases. In case of shoot, most of the key genes were identified in the saddlebrown module showing one unknown function protein.

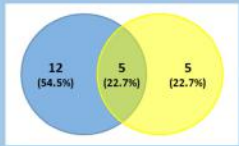


Figure 4: Venn diagram illustrating the common pathways overlapping enriched in root (blue circle) and shoot (yellow circle)

CONCLUSION

We provide for the first time organ-specific PPI networks of root and shoot by using WGCNA. The analysis of the module nodes by functional enrichment and the study of their topological features drive the identification of most enriched biological functions with the involvement of genes and key connectors. These results can be useful to focus systemically on the biology of *A. halleri* hypothesizing functions for unknown protein and/or driving experimental research.

REFERENCES

Corso M, et al. Adaptation of *Arabidopsis halleri* to extreme metal pollution through limited metal accumulation involves changes in cell wall composition and metal homeostasis. New Phytol. 2021

Chin CH, et al. Cytohubba: identifying hub objects and sub-networks from complex interactome. BMC Syst. Biol. 2014, 8, S11.

Ding Z & Kihara D. Computational identification of protein-protein interactions in model plant proteomes. Sci Rep. 2019, 9(1):8740.

Honjo MN & Kudoh H. Arabidopsis halleri: a perennial model system for studying population differentiation and local adaptation.

Karimizadeh E, et al. Analysis of gene expression profiles and protein-protein interaction networks in multiple tissues of systemic sclerosis. BMC Med Genomics. 2019, 12(1):199.

Langfelder P. & Horvath S. WGCNA: an R package for weighted correlation network analysis. BMC Bioinformatics, 2008, 9, S59.

Senkler J, et al. The mitochondrial complexome of Arabidopsis thaliana. Plant J. 2017, 89(6):1079-1092.